# NNUI_CELL

A Continuum Field Model for Self-Settling User Interfaces
Governed by a Rheological Constitutive Law and a Genotype–Phenotype Expression Layer

Proof of Concept & Formal Specification — v0.2 (post-validation revision)

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**Abstract**

Contemporary UI is authored as a large set of independent boxes, each carrying private geometry and style. Positions are hand-placed or solved by ad-hoc constraint systems, and adaptation to new content, new screens, or agent-driven edits requires re-authoring geometry. We propose **NNUI_CELL**, a model that replaces the box abstraction with a single *continuous medium* whose layout is never authored but always *solved* from local material state. The medium is governed by two interlocking formal axes. The **rheological axis** treats every region as yield-stress fluid governed by a solidity-modulated Herschel–Bulkley constitutive law with a rate-dependent yield and thixotropic settling; layout equilibrium ("settled") is defined as an energy minimum reached by overdamped gradient flow and terminated by a strain-rate threshold. The **genetic axis** treats every region as a cell carrying a *genome* (absorbed data traits) that deterministically *expresses* a *phenotype* (rendered controls and skin) through an extensible registry. The two axes share a single interface primitive: a boundary carries an always-on material-diffusion coefficient and an opt-in, deterministic gene-transfer operation. We give the constitutive derivations, the discrete real-time solver (XPBD with viscosity-modulated compliance), a construction for bit-identical determinism, the genetic operator set (injection, mitosis, symbiotic binding), a compact declarative grammar intended for LLM/agent emission, and seven executable invariants that pin the design against reductive drift. We close with four designed proof-of-concept experiments and an honest account of limitations. **Revision v0.2:** all four experiments have since been executed and passed on a headless prototype; this revision folds the validated engineering corrections into the discrete realization — most importantly a formal *arrest mechanism* (yield deadband, structure ratchet, jamming latch) that makes "settled" an exact, honestly measured state — and converts the experiment plan into results. The continuum theory of v0.1 is unchanged.

## Revision history

| | |
|---|---|
| **v0.1** | Specification with theory frozen; four designed experiments (R1–R4) stated with binary exit criteria; no implementation. |
| **v0.2** | Post-validation revision. R1–R4 executed and passed (§11); four validated corrections to the discrete realization admitted as normative (§6); the arrest mechanism formalized — yield deadband, structure ratchet, jamming latch, honest strain measurement (§7); Invariant 4 sharpened accordingly (§10); limitations rewritten against evidence. The constitutive model (§4.1–§4.6) and the genetic axis (§8) are *unchanged*: theory remains frozen; only its discrete realization is sharpened by measurement. |

# 1 Motivation: the box model is the wrong primitive

The dominant UI abstraction is a tree of independent rectangles. Each node owns its box, its style, and — through cascade, flexbox, or grid — a share of the negotiation over space. Three costs follow directly from the primitive itself, not from any particular framework:

1. **Assets clash.** Two components were never the same substance, so when they meet they overlap, collide, or must be manually reconciled. Integration is glue work.
2. **Layout precedes design.** The pipeline is scaffold → fit → lay out → *then* design. Structure and appearance are separate passes because boxes hold both independently.
3. **Adaptation re-authors geometry.** New content, a resized viewport, or an agent edit re-invokes an authoring step. For an autonomous agent this means emitting and iterating explicit coordinates — expensive in tokens and brittle under change.

NNUI_CELL removes the primitive. There are no independent boxes. There is one continuous medium, and a "component" is a region of that medium distinguished only by its local parameter field and its skin. Every consequence in this paper is a consequence of that single move; the rheology and the biology are the two mechanisms that make a continuum tractable and addressable.

## 1.1 Contributions

- A **solidity-modulated Herschel–Bulkley** constitutive law (§4.1) that folds "yield stress," "consistency," and "locked" onto a single continuous field $S \in [0, 1]$, so that locking is the top of the resistance dial rather than a separate mechanism.
- A **rate-dependent yield** formulation (§4.2) that resolves an apparent contradiction in the design intent — a material cannot be both shear-thinning and shear-thickening — by placing the rate dependence in the *yield*, not the flow index.
- A definition of **settling** as overdamped gradient flow of an explicit energy functional (§4.5), terminated by a measurable strain-rate threshold.
- **Viscosity-modulated XPBD compliance** (§5): the bridge that turns the continuum law into a stable 60 fps solver by setting each constraint's compliance from the local apparent viscosity. This is the engineering core.
- A deterministic **genotype→phenotype** expression layer (§8) with an extensible registry, and a reduction of the biological operations (injection, mitosis, binding) to defined, non-stochastic mechanisms.
- A **unified boundary operator** (§8.4) carrying two decoupled coefficients, so physical proximity never forces genetic fusion.
- Seven **executable invariants** (§10) that convert anti-drift intent into automated acceptance tests.

# 2 Positioning and prior art

We separate what exists from what we claim is new, to keep the novelty statement honest.

- **Spring / physics-flavoured animation** (iOS UIKit dynamics, Framer Motion, React Spring) animates *between* authored layouts using easing or spring presets. Layout itself is still box-authored, and the physics is decorative. NNUI_CELL has no animation system at all (Invariant 6); motion is the solver relaxing, never a tween.

- **Constraint layout** (Cassowary/AutoLayout, CSS flex/grid) solves geometry from declared relations. This is the closest neighbour. The difference is the governing law: constraint solvers optimize algebraic relations to a unique satisfying assignment; NNUI_CELL evolves a *material* under a rheological law toward an energy minimum, giving it settling dynamics, yield behaviour, locking, and content-preserving flow as one continuum rather than as separate features.
- **Soft-body / position-based dynamics** (PBD/XPBD in games) simulates deformable matter. We reuse the solver but drive its parameters from a rheological constitutive law and attach a semantic (genetic) layer, which PBD does not have.
- **Blob / metaball visual effects** are rendering tricks with no layout semantics.
- **Data-driven component selection** (schema-to-widget mappers, BI auto-charting) resembles our expression layer but is not embedded in a shared physical medium and has no notion of trait inheritance across a shared interface.

**Novelty claim (defensible):** *a layout engine whose constitutive law is an actual rheological equation, in which authors and agents parameterize material properties and containment rather than coordinates, unified with a deterministic genotype→phenotype expression layer at a single interface primitive.* We are unaware of prior art combining these.

# 3 The two-axis model

A region of the medium — a *cell* — is fully specified only by two orthogonal axes.

| **Axis 1 — Rheological**<br>*how it flows* | **Axis 2 — Genetic**<br>*what it becomes* |
|---|---|
| Geometry over time: position, motion, settling, locking, boundaries.<br>State: $S$, $n$, $\xi$ (per point); $D$, bind (per interface).<br>Mechanism: solidity-modulated Herschel–Bulkley; energy-minimizing settling. | Identity and manifestation: what data a cell holds and which controls it grows.<br>State: genome $\Gamma$; phenotype $\Phi = \text{express}(\Gamma)$.<br>Mechanism: sequencing seq; deterministic expression via registry $\mathcal{R}$. |

Design principle. The rheological axis alone cannot say how a dropped CSV becomes a chart; the genetic axis alone cannot say where the chart goes or how it settles. The model's expressive power is the product, not the sum, of the two axes.

# 4 Rheological axis: constitutive model

## 4.1 Solidity-modulated Herschel–Bulkley law

The classical Herschel–Bulkley (HB) model describes a material that is rigid below a yield stress $\tau_y$ and flows above it:

$$\tau = \begin{cases} \text{rigid}, & \tau \le \tau_y, \\ \tau_y + K\,\dot{\gamma}^{\,n}, & \tau > \tau_y, \end{cases} \tag{1}$$

with consistency $K$ and flow index $n$ ($n < 1$ shear-thinning, $n > 1$ shear-thickening), $\dot{\gamma}$ the shear rate.

We introduce a single *solidity* field $S(\mathbf{x}) \in [0, 1)$ that jointly modulates yield and consistency, so a single dial spans "freely flowing" to "locked":

$$\tau_y(S) = \tau_y^0 \left( \frac{S}{1-S} \right)^{\kappa}, \tag{2}$$

$$K(S) = K^0 \, (1 + \beta S), \tag{3}$$

with base scale $\tau_y^0$, shaping exponent $\kappa > 0$, and consistency gain $\beta \geq 0$. The pole at $S = 1$ in (2) is deliberate: $\tau_y \to \infty$ as $S \to 1$, so no finite input can yield the region — **locking is the limit** $S \to 1$, not a separate flag. This collapses the three lock tools into one primitive, `set_solidity(mask,S)` (§10.1).

For numerical use we adopt the Papanastasiou regularization of the yield term, which replaces the discontinuity by a steep smooth transition governed by a regularization parameter $m$:

$$\eta_{\text{app}}(\dot{\gamma}, S) \;=\; \tau_y(S) \, \frac{1 - e^{-m\dot{\gamma}}}{\dot{\gamma}} \;+\; K(S) \, \dot{\gamma}^{\,n-1}, \tag{4}$$

where $\eta_{\text{app}}$ is the apparent viscosity relating stress to rate, $\tau = \eta_{\text{app}} \dot{\gamma}$. Equation (4) is the quantity the solver consumes.

### 4.2 The dual-rate response: rate-dependent yield

The design intent contains an apparent contradiction. It asks for (a) rigidity by default; (b) a fast, accidental impulse to *stiffen* and snap (Oobleck-like, shear-thickening); and (c) a slow, deliberate press to *soften* and flow (reverse-Oobleck, shear-thinning). A single flow index $n$ cannot be both $< 1$ and $> 1$.

The resolution is to separate the two effects onto different terms. Post-yield flow is shear-thinning ($n < 1$: once you have committed, the material gets out of your way). The impulse-stiffening lives not in $n$ but in a *rate-dependent yield*:

$$\tau_y^{\text{eff}}(S, \xi, \dot{\gamma}) \;=\; \underbrace{\tau_y^0 \left( \tfrac{S}{1-S} \right)^{\kappa}}_{\text{solidity}} \cdot \underbrace{\xi}_{\text{structure}} \cdot \underbrace{\left(1 + (\dot{\gamma}/\dot{\gamma}_c)^q\right)}_{\text{impact hardening}}, \tag{5}$$

with critical rate $\dot{\gamma}_c$ and hardening exponent $q > 0$; $\xi$ is the thixotropic structure parameter of §4.3. A slow deliberate input ($\dot{\gamma} \ll \dot{\gamma}_c$) sees $\tau_y^{\text{eff}} \approx \tau_y(S)\xi$ and, if sustained, exceeds it and flows smoothly (thinning). A fast flick ($\dot{\gamma} \gg \dot{\gamma}_c$) sees the yield spike by $(\dot{\gamma}/\dot{\gamma}_c)^q$, so the impulse is rejected and the region reads as solid. This is a rate-dependent yield-stress fluid — real rheology — and it is the honest unification the flat model hid.

### 4.3 Thixotropic settling: the "ooze"

Settling requires viscosity to *rebuild* once input stops. We attach a structure parameter $\xi(\mathbf{x}, t) \in [0, 1]$ with standard breakdown/rebuild kinetics:

$$\frac{d\xi}{dt} \;=\; \frac{1}{T_r} \, (1 - \xi) \;-\; a \, \xi \, \dot{\gamma}, \tag{6}$$

rebuild timescale $T_r$, breakdown coefficient $a$. Under shear the second term dominates and structure erodes ($\xi \downarrow$, material softens and flows); at rest the first term restores $\xi \to 1$ over $\sim T_r$, raising the effective yield through (5) until motion falls below threshold. The visible "moves around, then settles into the bowl" is exactly this relaxation. $T_r$ is the "ooze" dial; the design notes' thixotropy dial $\lambda$ is this $\xi$ kinetics.

### 4.4 Content as rigid inclusions

Text and images must never shear when their region flows. We model each content payload as a *rigid inclusion* suspended in the yield-stress matrix: only the matrix deforms; the inclusion translates and rotates as a rigid body. Formally, inclusion $i$ contributes a deformation penalty (§4.5) enforced as a rigid shape-matching constraint in the solver (§5), guaranteeing a paragraph moves without smearing.

### 4.5 Settling as overdamped gradient flow

Let $\mathbf{q}$ be the medium's configuration (discretized in §5). The rest state is the minimizer of a layout energy

$$E(\mathbf{q}) \;=\; \underbrace{w_f \int_\Omega \big(1-\rho(\mathbf{x})\big)^2 d\mathbf{x}}_{E_{\text{fill}}} \;+\; \underbrace{w_c \sum_{(i,j)\in\partial} \sigma_{ij}\,\ell_{ij}}_{E_{\text{cohesion}}} \;+\; \underbrace{w_r \sum_i \big\|\mathbf{F}_i - \mathbf{R}_i\big\|_F^2}_{E_{\text{content}}}, \tag{7}$$

where $\rho$ is occupancy over the container $\Omega$ (so $E_{\text{fill}}$ penalizes voids — the slime fills the bowl); $\sigma_{ij}$ is interfacial tension along boundary segment of length $\ell_{ij}$ (so $E_{\text{cohesion}}$ minimizes interface length, producing coalescence and menisci); and for each inclusion $\mathbf{F}_i$ is the deformation gradient with $\mathbf{R}_i$ its best-fit rotation (so $E_{\text{content}}$ penalizes any non-rigid deformation of content). The weights $w_f, w_c, w_r$ are global design parameters.

The medium follows overdamped dynamics — inertia is dominated by the rheological dissipation, so we integrate a damped descent

$$\mathbf{M}\,\dot{\mathbf{q}} \;=\; -\nabla E(\mathbf{q}) \;-\; \mathbf{C}(\eta_{\text{app}})\,\dot{\mathbf{q}}, \tag{8}$$

with mass matrix $\mathbf{M}$ and viscous damping $\mathbf{C}$ set by the local apparent viscosity. **Settled** is a measurable predicate, not a duration:

$$\text{settled} \iff \|\dot{\gamma}\|_\infty < \varepsilon_{\text{settle}}. \tag{9}$$

Relaxation is asymptotic and never clamped; rendering is cut below a perceptual threshold. The "alive" quality is precisely this un-clamped decay — never a snap.

### 4.6 Assimilation as controlled diffusion

When a new region is injected, its material-parameter field $\phi = (S, n, \xi, \dots)$ blends into its neighbours by diffusion across the shared boundary:

$$\frac{\partial \phi}{\partial t} \;=\; \nabla\cdot\big(D(\mathbf{x})\,\nabla\phi\big), \tag{10}$$

with assimilation coefficient field $D$. Large $D$: the region melts in within a few frames. $D = 0$: it never blends — a permanent distinct island (modals, pinned widgets). Note (10) diffuses *material parameters*, not content; content is a rigid inclusion (§4.4) and does not diffuse. The discrete explicit scheme $\phi^{t+1} = \phi^t + \Delta t\, D\, \nabla_h^2 \phi^t$ is stable under the diffusion CFL condition $\Delta t \le h^2/(2d\,D)$ in $d$ dimensions with grid spacing $h$.

## 5 Discretization and the real-time solver

### 5.1 Why XPBD

Full Navier–Stokes or SPH would be accurate and unaffordable, and its results are not needed: we want stable, controllable, real-time relaxation, not physical fidelity. Extended Position-Based Dynamics (XPBD) provides unconditional stability, real-time performance on modest

hardware, and — critically — a per-constraint *compliance* that we can drive from the constitutive law. The medium is discretized as clusters of particles connected by constraints (distance, volume/occupancy, rigid shape-matching for inclusions, interface tension).

### 5.2 Viscosity-modulated compliance (the bridge)

XPBD solves each constraint $C_j$ with the update

$$\Delta\lambda_j \;=\; \frac{-\,C_j(\mathbf{x}) - \tilde{\alpha}_j\,\lambda_j}{\nabla C_j\,\mathbf{M}^{-1}\,\nabla C_j^{\top} + \tilde{\alpha}_j}, \qquad \Delta\mathbf{x} \;=\; \mathbf{M}^{-1}\nabla C_j^{\top}\,\Delta\lambda_j, \tag{11}$$

where $\lambda_j$ is the constraint's Lagrange multiplier and $\tilde{\alpha}_j = \alpha_j/\Delta t^2$ is its compliance. The central move of this paper is to set compliance from the local apparent viscosity (4):

$$\alpha_j \;=\; \frac{1}{k_0\, g\big(\eta_{\text{app}}(S,\xi,\dot{\gamma})\big)}, \qquad g \text{ monotone increasing.} \tag{12}$$

As $S \to 1$, $\eta_{\text{app}} \to \infty$, so $\alpha_j \to 0$ and the constraint becomes perfectly rigid — the region is *locked*. As $S \to 0$ the constraint is highly compliant — the region *flows*. Thus one field $S$, through (4)–(12), spans the entire range from liquid to locked, per constraint, per frame. Rigid content inclusions use $\alpha_j \to 0$ shape-matching constraints, realizing $E_{\text{content}}$.

### 5.3 The solver loop

**Algorithm 1:** One deterministic solver tick (v0.2: overdamped predict + arrest phase)

**Input:** particles $\mathbf{x}$, velocities $\mathbf{v}$, fields $(S, n, \xi, D)$, constraints $\{C_j\}$
**Output:** next-frame $\mathbf{x}', \mathbf{v}'$

$\mathbf{x}^p \leftarrow \mathbf{x} + c\,\Delta t\,\mathbf{v} + \Delta t^2\,\mathbf{b}$ `// overdamped predict: carry` $c{\approx}0.2$ `(C2); body/layout forces b enter as displacements`
estimate local shear rate $\dot{\gamma}$ from the predicted strain
update structure $\xi$ by (6); update $\eta_{\text{app}}$ by (4)
diffuse fields $\phi$ by (10) (assimilation)
**for** *iter* $= 1 \ldots N_{solve}$ **do**
  **foreach** *constraint* $C_j$ ***in fixed sorted order*** **do**
    compute compliance $\tilde{\alpha}_j$ by (12)
    apply XPBD update (11) to $\mathbf{x}^p$
  **end**
**end**
apply the **arrest phase** (Algorithm 2): yield deadband, structure ratchet, jamming latch
$\mathbf{v}' \leftarrow (\mathbf{x}^p - \mathbf{x})/\Delta t$ *after* arrest; $\mathbf{x}' \leftarrow \mathbf{x}^p$
**if** $\|\dot{\gamma}_{realized}\|_\infty < \varepsilon_{settle}$ **then**
  mark region **settled**; freeze rendering
**end**

v0.2 change: the predictor and the tail of the tick are revised per the validated corrections (§6) and the arrest mechanism (§7). Constraint projection itself, and the compliance bridge (12), are unchanged. Engineering note: any acceleration structure used for contact generation (spatial hash, grid) must be rebuilt *every projection iteration*, not once per tick — stale neighbourhoods miss drifted overlaps and manifest as a pile that never stabilizes.

### 5.4 Determinism by construction

Bit-identical settling (experiment R2) is required for testing, accessibility, and agent trust. It is achieved, not hoped for: (i) fixed timestep $\Delta t$; (ii) constraints solved in a fixed order sorted by stable identifier (Gauss–Seidel, no data races); (iii) all reductions performed in a fixed, sorted order so floating-point addition is associative-stable; (iv) no wall-clock or uninitialized state enters the update; (v) where cross-hardware reproducibility is needed, fixed-point or a fixed float evaluation order is mandated. Under these, identical inputs yield identical settled states on every run and machine.

## 6 Validated corrections to the discrete realization

R1–R4 (§11) were executed on a dependency-free headless prototype after v0.1. Four corrections emerged that are now *normative*: an implementation that ignores them reproduces the failures we measured. The continuum theory required no change; every correction concerns how the theory is realized discretely.

**C1. Non-penetration is stiff for every material.** Constraints partition into *material* constraints (cohesion, turgor, shape-matching), whose compliance is set from viscosity by (12), and *topological* constraints (inter-region contact, container walls), whose compliance is always near zero. Softness lives in internal deformation, never in whether matter may interpenetrate. Scaling contact compliance by material softness — v0.1's implied uniform reading of (12) — jams the solve.

**C2. "Overdamped" means little carried momentum.** The predictor carries a small fraction $c \approx 0.2$ of velocity, and body/layout forces enter as quasi-static displacements. Carrying most of the velocity (heavy *per-frame damping* of an otherwise inertial integrator) is under-damped in effect: measured, it produced exact deterministic limit cycles — perpetual slosh with the strain-rate readout frozen at a constant — rather than convergence to rest.

**C3. Thixotropy is load-bearing, not decorative.** A frictionless frustrated packing creeps forever; no tuning of damping alone arrests it. The $\xi$ kinetics of (6) — structure rebuilding at rest, raising the effective yield until sub-yield creep is arrested — *is* the physical settle. The metaphor turned out to be the mechanism. Its full discrete form is the arrest mechanism of §7.

**C4. Initial conditions are part of the contract.** Near-solution seeding (jittered lattice; spatially spread injection) settles fast and cleanly; pathological seeding (deeply interpenetrating spawn, jammed clump) costs an order of magnitude in settle time or worse. Spawn placement is a normative part of the injection operator, not a cosmetic choice.

## 7 The arrest mechanism: how "settled" becomes exact

v0.1 defined settling by threshold (9) over the smooth, Papanastasiou-regularized law. Validation exposed a gap: the regularized law is correct for the *flowing* branch, but a smooth law alone never produces *exact* rest in a discrete solver — gravity re-injects displacement every tick, Gauss–Seidel projection cancels it imperfectly, and the residual holds the medium in measurable perpetual micro-motion. The rigid branch of a yield-stress material must be realized *exactly*. Three interlocking mechanisms do so; all three were forced by measured failures, and all three are direct discrete realizations of the constitutive model.

### 7.1 Yield deadband: the no-flow branch

Below yield, a Bingham-class material does not flow at all; it sustains the stress elastically. Discretely: after constraint projection, a point whose realized step speed falls below the local yield velocity is *reverted to its pre-step position*,

$$\|\mathbf{x}_i^p - \mathbf{x}_i\| / \Delta t \;<\; v_y(S,\xi) \;\;\Longrightarrow\;\; \mathbf{x}_i^p \leftarrow \mathbf{x}_i, \qquad v_y(S,\xi) \;=\; v_0 \left(1 + c_\xi\, \xi\right) h\big(\tau_y(S)\big), \tag{13}$$

with $h$ monotone increasing. The material *holds the residual sub-yield stress at rest* — small unresolved constraint violations persist as internal prestress, exactly as a physical yield-stress medium sustains stress below $\tau_y$. Because $v_y$ grows with $\tau_y(S)$, the solidity dial *is* the yield: low $S$ pours, mid $S$ is paste, $S \to 1$ recovers the lock as the deadband diverges — consistent with (12), now from the flow side as well.

### 7.2 Structure ratchet: thixotropy closing the loop

The deadband alone leaves a band of slow creep just above $v_y$. The $\xi$ kinetics (6) close it: $\xi$ multiplies the yield in (5) and the deadband in (13), so matter that slows *gains* yield and is ratcheted to a full stop, while matter under sustained shear keeps $\xi$ low and flows freely. This is hysteresis by construction — the settled state is strictly harder to leave than to approach. One boundary condition is normative: while the containing frame itself accelerates (a slosh of the vessel; a drag of the whole canvas), the entire body is under shear and $\xi$ rebuild is suspended.

### 7.3 Jamming latch: caged rattle is rest

A discrete element deep in a jammed packing can rattle indefinitely between neighbours — per-tick corrections above $v_y$ with zero net transport — re-agitating its surroundings and preventing convergence (measured as settled-state flicker). Granular physics names this state: the element is *caged*, and a caged element is macroscopically at rest. Discretely: if $\mathbf{x}_i$ has remained within radius $r_{\text{cage}}$ of an anchor for time $T_{\text{cage}}$, set $\xi_i = 1$ and cease body-force injection for $i$; remaining constraint corrections then relax monotonically (nothing re-excites them). The latch is held while the cage condition holds and releases the moment the element genuinely leaves its cage. With deadband, ratchet, and latch together, the measured settled state is exact ($\|\dot\gamma\|_\infty = 0$) and stable (zero flicker over sustained observation).

### 7.4 Honest strain measurement

The settle predicate (9) is meaningful only if $\dot\gamma$ is measured on *realized* displacement, i.e. $\|(\mathbf{x}' - \mathbf{x})/\Delta t\|$ evaluated *after* the arrest phase, and never on a damped or zeroed velocity variable. Validation caught an instrument that reported *settled* while matter still moved (velocity zeroed by an arrest rule, positions still advancing under body force). The strain readout is part of the product's truthfulness, and Invariant 4 is sharpened accordingly (§10).

### 7.5 Disturbances must be physical

Under the overdamped predictor (C2), a velocity impulse decays by factor $c$ per tick — dead in $\mathcal{O}(1)$ frames, invisible. Admissible disturbances are therefore the physical ones: *sustained body force* (accelerating the containing frame — a slosh) and *direct displacement* (matter moved under the pointer, with local $\xi$ destroyed by the imposed shear). Both re-fluidize through the constitutive law itself; neither requires, nor permits, an animation-system side channel (Invariant 6).

**Algorithm 2:** Arrest phase (per tick, after constraint projection)

**Input:** pre-step $\mathbf{x}$, post-projection $\mathbf{x}^p$, structure $\xi$, fields $(S, \dots)$

**foreach** *point* $i$ **do**
  $s_i \leftarrow \|\mathbf{x}_i^p - \mathbf{x}_i\| / \Delta t$
  **if** $s_i < v_y(S_i, \xi_i)$ (13) **then** $\mathbf{x}_i^p \leftarrow \mathbf{x}_i$;
  $s_i \leftarrow 0$
  update $\xi_i$ by (6) with $\dot{\gamma} = s_i$ (rebuild suspended while frame accelerates)
  **if** *i caged* ($\|\mathbf{x}_i^p - \mathbf{a}_i\| < r_{cage}$ *for* $T_{cage}$) **then** $\xi_i \leftarrow 1$
  **else if** $\|\mathbf{x}_i^p - \mathbf{a}_i\| \geq r_{cage}$ **then**
    reset anchor $\mathbf{a}_i \leftarrow \mathbf{x}_i^p$
  **end**
**end**
report $\|\dot{\gamma}_{\text{realized}}\|_\infty = \max_i s_i$ `// honest strain: post-arrest only`

# 8 Genetic axis: identity and expression

## 8.1 Genome, germline, sequencing

A cell carries a *genome* $\Gamma$, a set of typed *traits* (capability tokens such as `tabular`, `temporal`, `media:video`, `numeric-range`). Every cell is born with a *germline* $\Gamma_0$ expressing primitive phenotypes (bare panel, container, control surface); **there is no empty cell**, only a cell whose genome is $\Gamma_0$. This removes any special case for "empty" and makes the shipped default UI the same substance as user-created UI. Absorbing an asset *adds* genes:

$$\text{seq} : \text{Bytes} \to 2^{\text{Trait}}, \qquad \Gamma \;\leftarrow\; \Gamma \cup \text{seq}(\text{asset}). \tag{14}$$

seq is a pure function: MIME/structure/column-type detection in, trait-set out. No physics, no ambiguity, fully testable.

## 8.2 Expression as a deterministic registry map

Phenotype is a deterministic function of genome through an extensible registry $\mathcal{R}$, a partial map from trait signatures to phenotype constructors:

$$\Phi \;=\; \text{express}(\Gamma) \;=\; \text{compose}\left\{\, \mathcal{R}(\text{sig}) \;:\; \text{sig} \in \text{signatures}(\Gamma) \,\right\}. \tag{15}$$

A tabular genome expresses a chart phenotype; a video genome expresses a player phenotype. Extensibility is the whole plug-and-play claim made concrete: a new data type is a new entry in $\mathcal{R}$, added *without touching the core* (experiment R4). Expression is deterministic — there is no stochastic or mutation operator anywhere in the model.

## 8.3 Operations reduced to mechanism

Per Invariant 7, each biological term is admitted only after reduction to a defined mechanism.

**Algorithm 3:** Genetic operators

**Injection** inject(cell, asset):
$\Gamma \leftarrow \Gamma \cup \text{seq}(\text{asset})$; $\Phi \leftarrow \text{express}(\Gamma)$
set assimilation field $D$ on the new boundary; rheology renders the blend via (10)

**Mitosis** mitose(cell):
$\Gamma_{\text{child}} \leftarrow \text{DEEP-COPY}(\Gamma_{\text{parent}})$ `// identity copied`
child inherits **no** position or settled geometry
$\Phi_{\text{child}} \leftarrow \text{express}(\Gamma_{\text{child}})$; child re-settles under its own local pressure
division visual = one mass pinching in two under interface tension (free from §4.5)

**Binding** bind$(A, B)$ `// deliberate, deterministic`
on the marked shared interface: $\Gamma_A \leftarrow \Gamma_A \cup \tau(\Gamma_B)$, $\tau$ a deterministic (possibly partial) transfer
adjacency alone **never** triggers this

Mitosis is not "make a copy." Taken literally it is genome replication plus independent re-expression: a duplicated chart dropped into a cramped corner settles compact while its parent stays wide — same genotype, different phenotype-in-context. The naive "duplicate" verb would have hidden this correct behaviour.

### 8.4 The unified boundary operator

A boundary between two cells carries two *decoupled* coefficients:

| Coefficient | Governs | Default |
|---|---|---|
| $D$ (material diffusion) | do the regions blend physical material | always $> 0$ (one medium) |
| bind (gene transfer) | do the regions exchange genome | **off**; opt-in only |

This preserves the elegance of "one interface, two jobs" while ensuring physical proximity during settling never forces genetic fusion. Symbiotic binding is precisely the deliberate act of engaging the second coefficient. There is no ambient, proximity-driven, or random gene exchange: such behaviour would violate determinism (R2) and Invariant 7 and is explicitly excluded.

## 9 The agent interface: declarative grammar and token economy

The end goal is an interface an LLM/agent emits cheaply. An agent declares *material intent and containment*, not coordinates; the solver produces and maintains geometry. The grammar has two verb families — rheological and genetic — over one region model.

```
program := { region | op } ;
region := "cell" id "{" material content "}" ;
material := "S:" num "n:" num "xi:" num "D:" num [ "skin:" id ] ;
content := "hold" asset-ref | "germline" ;
op := "inject" id asset-ref
          | "solidity" mask num // t1/t2/t3 all reduce here
          | "mitose" id
          | "bind" id id
          | "contain" id "in" region-ref ;
mask := "all" | id | "not" id ; // whole / selection / complement
```

**Worked example (dashboard).** An agent asks for a wide revenue panel that fills remaining space with three cohesive KPI cells grouped beside it:

```
cell main { S:0.3 n:0.8 xi:1 D:0.6 hold revenue.csv } // expresses -> chart
cell k1 { S:0.5 n:0.8 xi:1 D:0.4 hold kpi1 }
```

```
cell k2 { S:0.5 n:0.8 xi:1 D:0.4 hold kpi2 }
cell k3 { S:0.5 n:0.8 xi:1 D:0.4 hold kpi3 }
bind k1 k2 ; bind k2 k3 // cohesive group
contain { main k1 k2 k3 } in viewport
```

No coordinates appear. Positions, sizes, the chart type, and the group packing are *solved*. When `revenue.csv` changes or the viewport resizes, the medium re-settles with *no further agent call* — the zero-cost-adaptation claim. The economic argument is therefore not merely fewer tokens at generation ($\mathcal{O}(\text{dials} + \text{containment})$ versus $\mathcal{O}(\text{explicit geometry})$) but the elimination of the re-authoring round-trip on every change.

v0.2 status: measured. R3 (§11) tokenized the three reference layouts with the cl100k_base tokenizer: 68 / 47 / 52 tokens — all under the 80-token criterion — and 2.43× cheaper overall than equivalent compact markup for the same layout intent. The measured saving is the *floor*; the structural claim — zero re-authoring on change — is additional.

## 10 Invariants as executable specification

Anti-drift is enforced by artifacts, not intentions. Each invariant is stated as a predicate that becomes an automated acceptance test; a build that fails any is, by definition, drift.

**I1. No authored positions.** No code path assigns a coordinate directly; every position is a solver output. *Test:* static check + perturbation test (positions change iff material state changes).

**I2. Real constitutive law.** $\eta_{\text{app}}$ is evaluated per frame from (4); changing $n$ at runtime changes observed behaviour. *Test:* sweep $n$, assert response monotonicity; assert no easing/spring preset in the motion path.

**I3. Locks are solidity.** No boolean lock flag; all locking flows through `set_solidity`. *Test:* grep/AST for lock special-casing returns empty; $S = 1$ region is unyielding under bounded input.

**I4. Threshold-terminated settling, honestly measured** (sharpened in v0.2). Settling ends by (9), never by a timer, and the strain rate in the predicate is the *realized post-arrest displacement rate* (§7) — never a damped or zeroed velocity variable. *Test:* varying content mass changes settle time; no duration constant gates it; an instrumented run must show the settled state is exact and flicker-free over a sustained observation window, and must show non-zero strain whenever positions change.

**I5. One medium.** No component owns private layout logic outside the solver. *Test:* removing a component type does not remove any layout code path.

**I6. No animation system.** The renderer is a stateless view of solver state; no tweens, timelines, or durations exist. *Test:* freezing the solver freezes all on-screen motion exactly.

**I7. Metaphor reduces to mechanism.** Every biological term in the codebase maps to a defined operator (§8); expression and all operators are deterministic. *Test:* R2 bit-identity holds; no stochastic call in the genetic path.

### 10.1 Tool primitives

The three authoring tools are not three features; they are three masks over one primitive:

| Tool | Meaning | Reduction |
|---|---|---|
| `slime-lock` (t1) | lock whole UI solid | `set_solidity(all, 1)` |
| `window-lock` (t2) | lock a selected area (inside frozen) | `set_solidity(sel, 1)` |
| `isolation` (t3) | work inside a region (outside frozen) | `set_solidity(not sel, 1)` |

Partial locks ($S = 0.8$: stiff but yielding to determined force) and soft gradient lock edges fall out for free, since $S$ is a continuous field.

## 11 Proof-of-concept plan: designed experiments

Theory was frozen at v0.1; the four experiments below were then executed on a dependency-free headless Rust prototype (std-only, single-threaded for determinism by construction, §5.4). **All four passed.** Exit criteria are restated with measured results.

| ID | Question | Exit criterion |
|---|---|---|
| R1 | Discretization: does XPBD (§5) settle at real-time scale? | **PASS.** 50 regions (1250 particles) settle at 1.2–1.4 ms/frame (budget 16); scaling holds to 400 regions / 10 000 particles at 11.8 ms. The four corrections of §6 were discovered here. |
| R2 | Determinism (§5.4). | **PASS** (within machine). 100 independent settle runs bit-identical (FNV-1a `0xdb5295e7d7bce689`); identical across separate process invocations. The two-machine clause remains untested (§12). |
| R3 | Agent grammar / token economy (§9). | **PASS.** Grammar lowers end-to-end to the R1 solver (parse → settle → hash), so "settles unambiguously" is demonstrated, not asserted. Dashboard/article/settings: 68/47/52 tokens (cl100k_base), all $< 80$; 2.43× cheaper than equivalent compact markup. |
| R4 | Expression registry (§8). | **PASS.** CSV→chart, MP4→player, TXT→reader; a fourth type (PNG→image viewer) added through the public registry with the core untouched — both detection and expression are registry-driven, so "no core change" is structural. |

The full suite is 15 automated tests (rheology, invariants, grammar, genetics), all green; the packaged crate rebuilds standalone and reproduces identical hashes.

**Corroboration — workstream B visual bench.** Separately from the headless gates, a browser-based fluid-in-vessel bench (one Herschel–Bulkley medium poured into a bowl, rendered as a single implicit surface) exercised the full arrest mechanism of §7 under instrumented observation: pours settle in ∼4–5 s to *exact* zero strain; a vessel slosh visibly surges and resettles in under a second; at 1200 particles the settled state showed *zero* flicker over a 10 s observation window at 8–11 ms/frame. Its first version — discrete glowing regions that never took the vessel's shape — was rejected as contradicting the continuum thesis and is retained in the project record as the canonical example of the drift Invariants 1–6 exist to catch; the rebuilt bench was verified by screenshot capture and programmatic image analysis (single connected

component spanning the vessel in every state). The bench illustrates; the crate remains the reference of record.

**Deferred by design** (stage-5 questions, excluded to prevent scope drift): OS integration, multi-user editing, CSS interop, and performance beyond the measured scale. Pipeline position: specification → R1–R4 prototype *(done)* → visual demo *(done)* → agent-facing API (workstream A, the goal) → integration.

## 12 Limitations and threats to validity

We state these plainly; a PoC that hides them is not professional.

- **Prototype scale, not product scale.** R1–R4 validate the discrete realization on a headless prototype (5×5-particle region blobs, no true rigid content inclusions yet). $E_{\text{content}}$ shape-matching and the inclusion pipeline remain specified but unexercised.
- **Parameters are tuned, not derived.** The measured operating point ($N_{\text{solve}}$, carry $c$, deadband constants $v_0, c_\xi$, cage constants $r_{\text{cage}}, T_{\text{cage}}$, regularization $m$) was reached by instrumented iteration against the exit criteria. The corrections of §6 are qualitative and forced; the constants are not.
- **Cross-hardware float determinism remains untested.** R2 passed within one machine and across processes; (11) over IEEE floats can still diverge across architectures, and the two-machine clause may force fixed-point with unquantified cost.
- **Token economy is measured narrowly.** 2.43× on three layouts against compact equivalent markup, with both sides using their own defaults. Layouts demanding genuine explicit control may narrow the advantage; the zero-re-authoring claim, while structural, is not yet quantified in tokens saved over a session.
- **The visual bench is an illustration.** Its fluid dynamics (gravity magnitude, deadband base) are tuned for a vessel-filling demonstration and differ numerically from the crate. Constitutive *form* is shared; constants are not. The crate is the reference.
- **Accessibility is an open design, but a favourable one.** A continuous medium has no DOM semantics for assistive tech — however, the *genome* is a clean semantic layer (traits and expressed phenotype), arguably a better accessibility source than inferred box structure. This must be designed, not assumed.
- **Continuum cost at scale.** Measured to 10 000 particles single-threaded; very large interfaces beyond this are unaddressed and deliberately deferred.

## 13 Conclusion

NNUI_CELL proposes a single structural break — *replace the box model with a continuum field* — and supplies the two mechanisms that make a continuum both movable and addressable: a solidity-modulated, rate-dependent-yield Herschel–Bulkley rheology that settles by energy minimization and locks at the top of one dial, and a deterministic genotype→phenotype expression layer that lets a region digest and manifest arbitrary data. The two axes meet at one interface primitive with decoupled physical and genetic coupling, so proximity never forces fusion and every biological term reduces to a defined, non-stochastic mechanism. The design is pinned against reductive drift by seven executable invariants. v0.1 stated its open questions as four designed experiments; v0.2 reports all four passed, and folds what implementation forced — stiff topology, an overdamped quasi-static integrator, and above all the arrest mechanism that makes

"settled" an exact, honestly measured, thixotropically latched state — back into the specification as normative. The theory did not change under contact with implementation; its discrete realization sharpened. What remains is not validation but construction: the agent-facing API that is the point of the system.

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*MIVIM Development — Standards in Novel Design.* Specification v0.2 (post-validation revision), July 2026; supersedes v0.1. Technical architecture developed in an AI-assisted research collaboration; all design authority and creative direction held by the author. R1–R4 results reported herein were produced by the accompanying headless prototype crate and are reproducible from its shipped sources.

# A Symbol and dial reference

| Symbol | Meaning | Role |
|---|---|---|
| $S \in [0, 1)$ | solidity field | dial: liquid→locked; per point |
| $n$ | flow index | dial: post-yield thinning ($< 1$) / thickening ($> 1$) |
| $\xi \in [0, 1]$ | thixotropic structure | state: erodes under shear, rebuilds at rest |
| $T_r$ | rebuild timescale | dial: "ooze" / settle speed |
| $D$ | assimilation coefficient | dial: melt-in ($> 0$) vs. island (0); per interface |
| $q, \dot{\gamma}_c$ | hardening exponent, critical rate | tune fast-flick rigidity |
| $\kappa, \beta$ | yield shaping, consistency gain | calibrate the $S$ curve |
| $w_f, w_c, w_r$ | energy weights | fill vs. cohesion vs. content rigidity |
| $\varepsilon_{\text{settle}}$ | strain-rate threshold | defines "settled" |
| $\alpha_j, \tilde{\alpha}_j$ | XPBD compliance | from $\eta_{\text{app}}$; material only (C1) |
| $c$ | momentum carry | predictor carry, $\approx 0.2$ (C2) |
| $v_0, c_\xi$ | deadband base, ratchet gain | yield deadband $v_y$, (13) |
| $r_{\text{cage}}, T_{\text{cage}}$ | cage radius, cage time | jamming latch (§7) |
| $\Gamma, \Gamma_0, \Phi$ | genome, germline, phenotype | genetic state |
| $\mathcal{R}$ | expression registry | extensible trait→phenotype map |

The full per-point material state is $\{S, n, \xi, \text{skin}\}$; the per-interface state is $\{D, \text{bind}\}$. This compactness is the token-economy argument: a handful of scalars and a containment relation describe what box markup approximates in hundreds of declarations.